

PUBLIC
EXHIBIT 53

U.S. Patent No. 9,865,546 Domestic Industry Chart

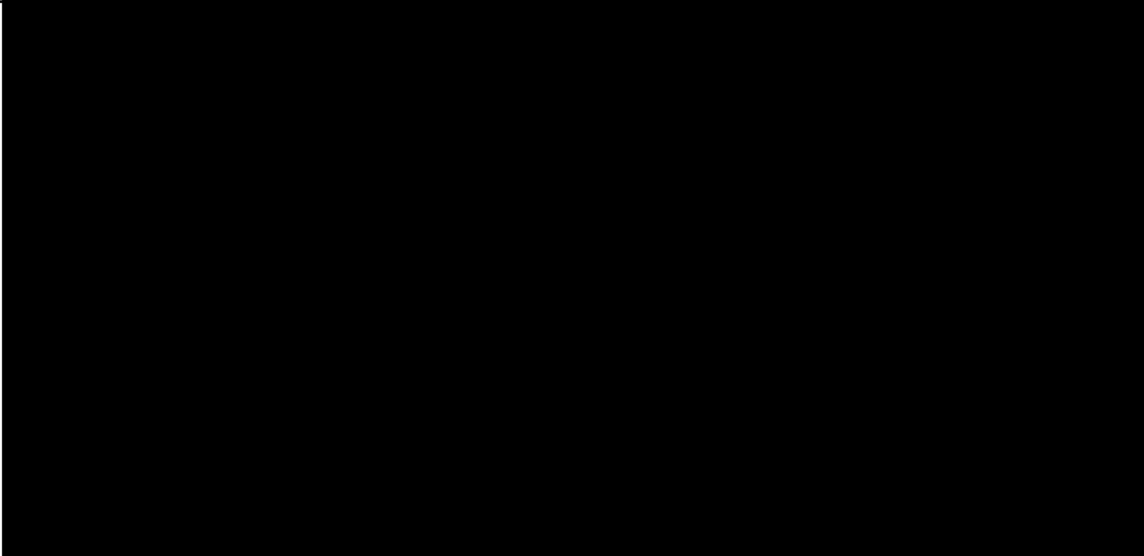
Title: Contacts to semiconductor substrate and methods of forming same

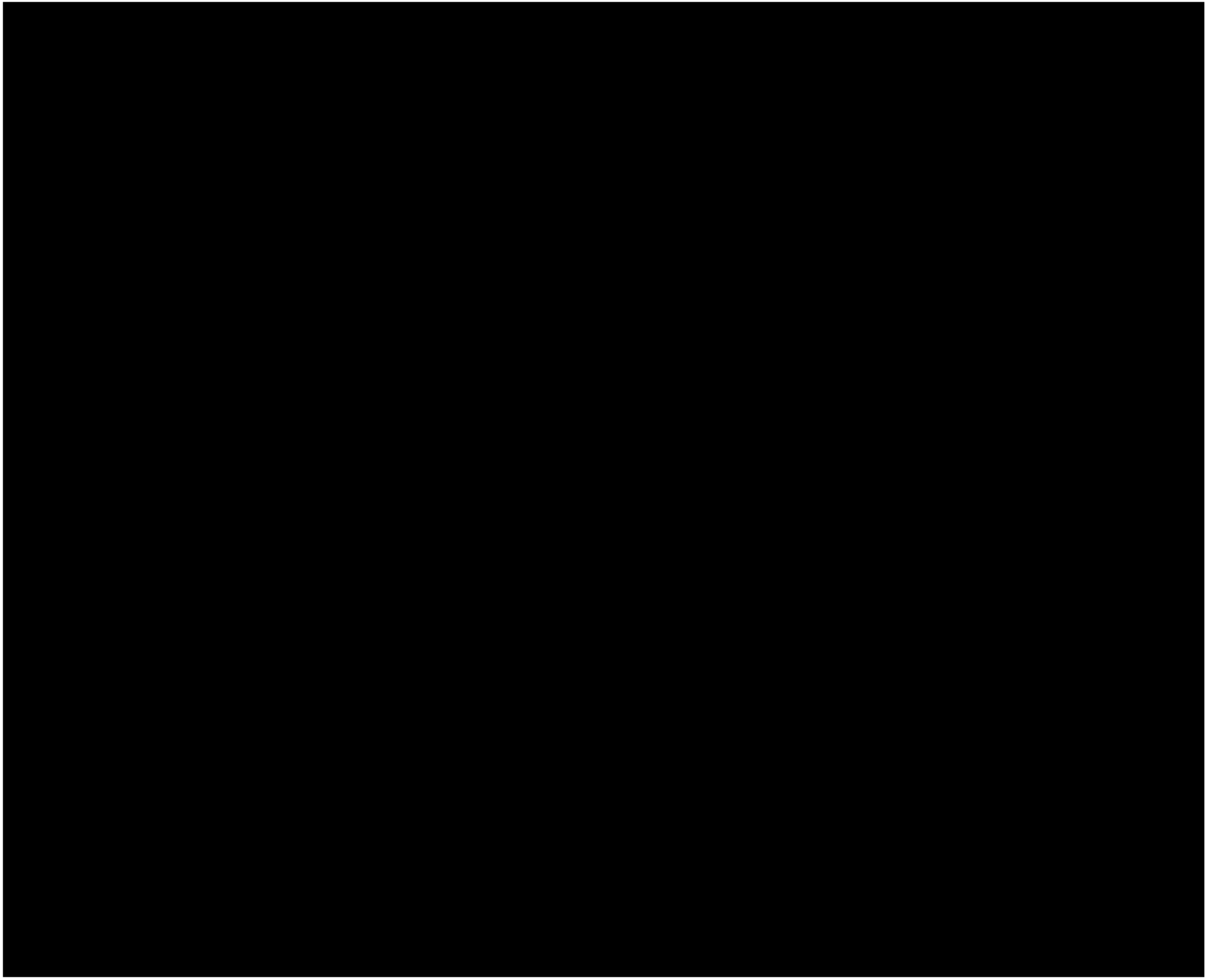
Inventors: Emre Alptekin, Nicolas L. Breil, Christian Lavoie, Ahmet S. Ozcan, Kathryn T. Schonenberg

Abstract: An aspect of the invention includes a method for forming a contact in a dielectric layer over a semiconductor substrate. The method may comprise: forming a contact opening in a dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate; depositing a first liner layer to conformally coat the contact opening; causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate; depositing a refractory metal layer over the first intermix region; and depositing a metal in the contact opening thereby forming the contact.

Claim	'546 Domestic Industry Process
[1pre] A method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising:	To the extent the preamble is limiting, the method used to manufacture chips using GlobalFoundries' 130CBIC Process (hereinafter, "'546 Domestic Industry Process") ¹ comprises a method for forming a contact in a dielectric layer to a semiconductor substrate.

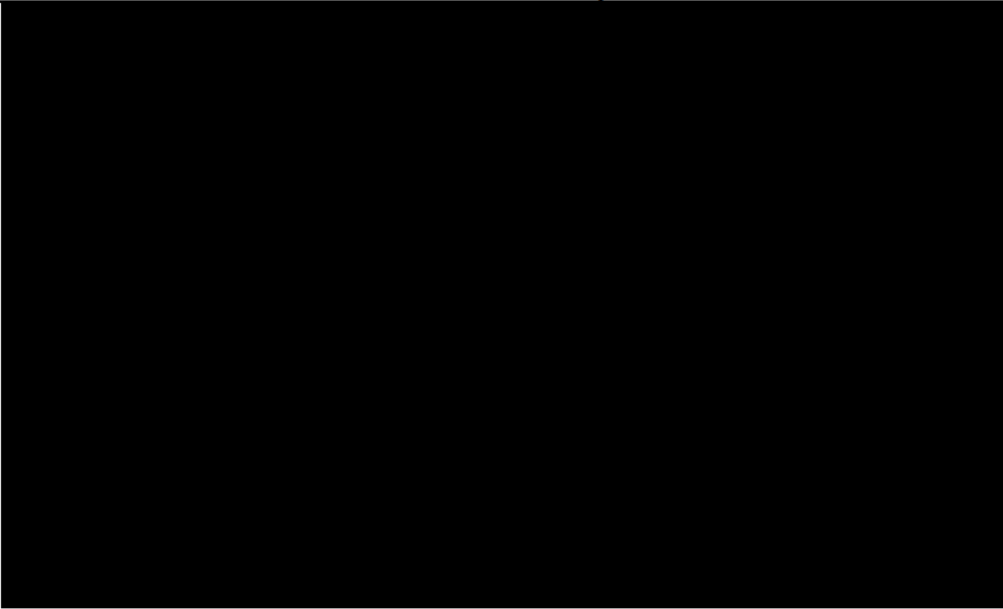
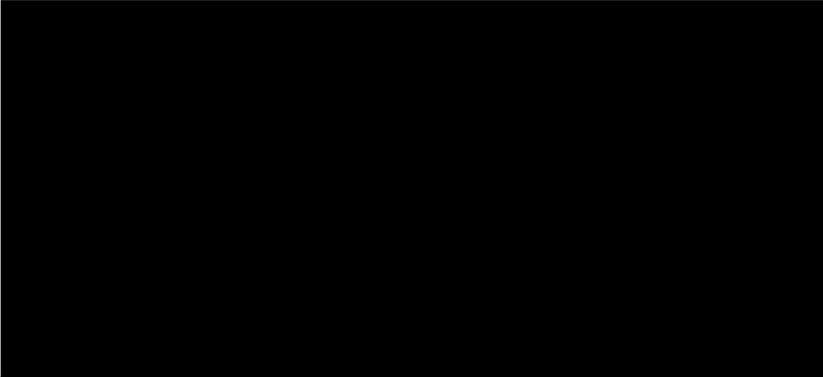
¹ GlobalFoundries reserves the right to identify different or additional '546 Domestic Industry Processes and amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances.

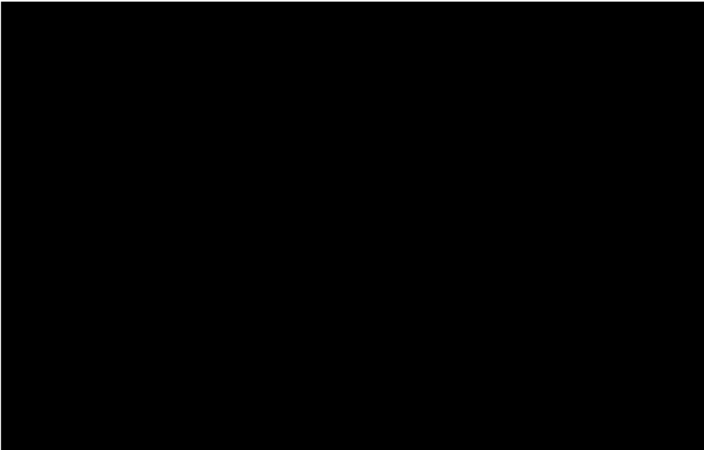
Claim	'546 Domestic Industry Process
	
[1a] forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate;	The '546 Domestic Industry Process comprises forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate. For example, the TEM reverse engineering analysis below shows a contact opening (after it has been filled) in a dielectric layer formed to expose an upper portion of the semiconductor substrate.

Claim	'546 Domestic Industry Process
	
[1b] depositing a first liner layer to conformally coat the contact opening;	The '546 Domestic Industry Process comprises depositing a first liner layer to conformally coat the contact opening.

Claim	'546 Domestic Industry Process
	For example, the '546 Domestic Industry Process includes depositing the first liner layer to conformally coat the contact opening. The presence of the first liner layer is shown in the materials analysis below. The first liner layer substantially coating the sidewalls of the contact opening can be seen more clearly in the material analysis below, which separates the elements above.

Claim	'546 Domestic Industry Process
[1c] causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate;	The '546 Domestic Industry Process comprises causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate. For example, the '546 Domestic Industry Process comprises formation of an intermix region at an upper portion of the semiconductor substrate, as shown in the TEM reverse engineering analysis below. The green arrow in the image immediately below corresponds to the green image at the top of the ensuing material analysis chart 

Claim	'546 Domestic Industry Process
	 For example, the images below shows the material stack for the source/drain region. The material analysis charts below show the composition of the source-side material stack. The green arrow pointing upwards in the figure below corresponds to the green arrow pointing from right to left in the following figure. 

Claim	'546 Domestic Industry Process
	
[1d] depositing a refractory metal layer over the first intermix region; and	The '546 Domestic Industry Process comprises depositing a refractory metal layer over the first intermix region. For example, the '546 Domestic Industry Process comprises depositing a refractory metal layer over the first intermix region, as shown in the TEM reverse engineering analysis below. 
[1e] depositing a metal in the contact opening thereby forming the contact.	The '546 Domestic Industry Process comprises depositing a metal in the contact opening thereby forming the contact.

Claim	'546 Domestic Industry Process
	